Zhuoyang Zou

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Research Interests

- **Agentic AI for Science and Healthcare:** multi-agent LLM systems for AI-assisted peer review, clinical decision support, and disease prediction.
- **LLM Reasoning and Agent Design:** adaptive planning, multi-agent debate, tool use, and long-horizon task execution.
- **Evaluation and Trustworthy AI:** benchmarks and detection methods that reveal when models rely on stylistic or spurious shortcuts instead of genuine reasoning.

Education

Pennsylvania State University Sep. 2024 – Present
Ph.D. in Computer Science, Advisor: Prof. Wenpeng Yin *University Park, PA*

Northwestern University Sep. 2022 – Dec. 2023
M.S. in Artificial Intelligence *Evanston, IL*

The Chinese University of Hong Kong Sep. 2017 – May 2022
B.S. in Computer Science, Advisor: Prof. Qi Dou *Hong Kong*

Honors and Awards

- **ICDS Rising Researcher Award**, Penn State Institute for Computational and Data Sciences 2026
- **Workday AI Ph.D. Fellowship** ($100,000 grant) 2026
- **NSF Research Traineeship Fellow**, LinDiv: Linguistic Diversity Across the Lifespan (#2125865) 2026 – 2027
- **Workshop Best Paper Award**, ICML 2025, for *AAAR-1.0* 2025

Publications

- **Z. Zou**, A. Ansari, J. Yang, D.C. Zhang, Q. Chen, D. Lee, W. Yin. **Who Wrote It Is Not Enough: Detecting Who Contributed the Insight.** *arXiv:2610.07365*, 2026. [arXiv]
- **Z. Zou**, A. Ansari, D.C. Zhang, D. Lee, W. Yin. **DIAGPaper: Diagnosing Valid and Specific Weaknesses in Scientific Papers via Multi-Agent Reasoning.** *arXiv:2601.07611*, 2026. [arXiv]
- J. Ahn, R. Kamoi, B. Atil, R. Lou, W. Kang, H. Park, S. Das, **Z. Zou**, et al. **Bridging the Know-Act Gap via Task-Level Autoregressive Reasoning.** *arXiv:2603.22619*, 2026. [arXiv]
- A. Ansari, D.C. Zhang, **Z. Zou**, W. Yin, D. Lee. **M2-Verify: A Large-Scale Multidomain Benchmark for Checking Multimodal Claim Consistency.** *arXiv:2604.01306*, 2026. [arXiv]
- B. Belland, W. Yin, **Z. Zou**, J. Mussoline, C. Kim. **AI-Enabled Automated Scaffolding for Undergraduate Students' Learning to Debug and Reason.** *AI4HE @ CSEDU*, 2026. [PDF]
- R. Lou, H. Xu, S. Wang, J. Du, R. Kamoi, X. Lu, J. Xie, Y. Sun, Y. Zhang, J. Ahn, **Z. Zou**, et al. **AAAR-1.0: Assessing AI's Potential to Assist Research.** *ICML Workshop*, 2025 **(Best Paper Award)**. [arXiv]
- Y. Zhang, W. Zheng, A. Madasu, P. Shi, R. Kamoi, H. Zhou, **Z. Zou**, et al. **HRScene: How Far Are VLMs from Effective High-Resolution Image Understanding?** *ICCV*, 2025. [arXiv]
- **Z. Zou**, P.T. Teo, A. Yalamanchili, M. Abazeed. **Optimizing Deep Learning Models: CNN-RNN Augmentation with Grad-CAM Analysis for Predicting Pneumonitis.** *Int. J. Radiation Oncology, Biology, Physics*, 2024. [Paper]

Research Experience

Pennsylvania State University Sep. 2024 – Present
Graduate Researcher, Agentic AI & Multi-Agent LLM Systems *University Park, PA*

- **InsightProv** *[Qwen2.5-72B, LoRA, adversarial training]*: Insight-provenance detection for scientific reviews: identifies whether a review's insights come from humans, LLMs, or both. Built **InsightProv-v0** (12,000+ reviews on 4,000+ papers, sentence-level labels); a two-stage adversarial framework unlearns stylistic authorship shortcuts. Found that AI insights stay close to generic or paper-provided content, while human insights add external knowledge and independent judgment.
- **DIAGPaper** *[Python, GPT-4o, Claude, Llama, LangChain]*: Multi-agent framework for automated peer review. A Customizer generates paper-specific criteria; a Rebuttal module runs author–reviewer debate, filtering 40–60% of invalid critiques; a Prioritizer, learned from 15,343 real reviews, ranks weaknesses by severity. **F1 51.89 on AAAR, 9.4% above GPT-4o.**
- **AAAR-1.0** *[Python, benchmark design]*: Large-scale benchmark for AI-assisted research. **ICML 2025 Workshop Best Paper Award.**

- **M2-Verify** *[VLMs, multimodal reasoning]*: Multidomain benchmark for claim consistency across text and images.
- **Know-Act Gap** *[LLM agents, autoregressive reasoning]*: Task-level autoregressive reasoning that turns LLM knowledge into executable actions in agentic settings.

Northwestern University Medical School Jul. 2023 – May 2024
Researcher, Medical AI *Chicago, IL*

- **iTox** *[PyTorch, CNN-RNN, Grad-CAM]*: Pneumonitis risk prediction from pre-treatment CT scans, **AUC 0.84**. Grad-CAM localizes predictive lung regions; the iTox dose metric reduces pneumonitis risk by **20%**.

The Chinese University of Hong Kong Jan. 2022 – Aug. 2022
Researcher, VR & Surgical AI *Hong Kong*

- **VR Surgical Training** *[Unity C#, stereoscopic rendering]*: Simulated da Vinci surgical procedures with stereoscopic vision for trainee surgeons.
- **Surgical Instrument Segmentation** *[U-Net, PyTorch]*: Segmentation and activity recognition on surgical video; eye tracking for adaptive training.

Industry Experience

High Fashion Group May 2024 – Aug. 2024
AI Specialist *Hong Kong*

- **Fabric Recognition** *[PyTorch, Siamese Network, ResNet]*: Few-shot image similarity model for fabric pattern matching and quality control.
- **Semantic Fabric Search** *[LLM, RAG, vector embeddings]*: Text-to-fabric retrieval using LLM query encoding and dense retrieval over the product database.
- **Trend Forecasting** *[RAG, LLM, structured data]*: Multi-source pipeline for fashion trend analysis and cost-performance reporting.

Karmo International Company Ltd. Jun. 2022 – May 2023
AI Developer *Hong Kong*

- **Construction Safety Detection** *[YOLOv5/v8, OpenCV]*: Real-time safety gear detection deployed at active construction sites.
- **Autonomous Vehicle Detection** *[TensorFlow]*: Multi-class real-time object detection deployed at Hong Kong Science Park.
- **Drainage Prediction** *[Django, Random Forest, scikit-learn]*: Web app and predictive model for the Hong Kong Drainage Services Department.

Technical Skills

- **LLMs & Agents:** multi-agent orchestration, tool use, RAG, LoRA fine-tuning, prompt engineering, LangChain, LangGraph, AutoGen
- **ML & Training:** PyTorch, HuggingFace Transformers, DeepSpeed ZeRO-3, multi-GPU training, TensorFlow, scikit-learn, Weights & Biases
- **Evaluation & Data:** benchmark design, experiment design, statistical testing, Pandas, NumPy, SQL, FAISS, ChromaDB
- **Engineering:** Python, C#, Docker, Git, Linux, AWS, FastAPI, Django, Claude Code, Cursor